

$$1.) (a) q = -k \frac{\Delta T}{L} \Rightarrow -237 \frac{W}{m \cdot K} \left( \frac{290-310}{400 \times 10^{-6} m} \right) = 1185000 \frac{W}{m^2}$$

$$\Rightarrow -237 \frac{W}{m \cdot K} \left( \frac{290-310}{400 \times 10^{-9} m} \right) = 1.185 \times 10^5 \frac{W}{m^2}$$

$$(b) q = -k \frac{\Delta T}{L} \Rightarrow -0.0263 \frac{W}{m \cdot K} \left( \frac{290-310}{400 \times 10^{-6} m} \right) = 1315 \frac{W}{m^2}$$

$$\Rightarrow -0.0263 \frac{W}{m \cdot K} \left( \frac{290-310}{400 \times 10^{-9} m} \right) = 1315000 \frac{W}{m^2}$$

$$(c) L_{A1} \Rightarrow (2 \times 40) \overset{\rightarrow S}{=} 80 nm$$

$$L_{air} \Rightarrow 400 - 80 = 320 nm$$

$$\Rightarrow -237 \left( \frac{290-310}{80 \times 10^{-6}} \right) = 59250000 \frac{W}{m^2}$$

$$\Rightarrow -0.0263 \left( \frac{290-310}{320 \times 10^{-6}} \right) = 1643.75 \frac{W}{m^2}$$

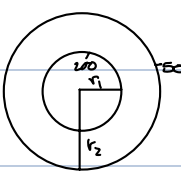
$$q''_{tot} = 59250000 + 1643.75 = 59251643.75 \frac{W}{m^2}$$

$$\Rightarrow -237 \left( \frac{290-310}{80 \times 10^{-9}} \right) = 5925 \times 10^{10} \frac{W}{m^2}$$

$$\Rightarrow -0.0263 \left( \frac{290-310}{320 \times 10^{-9}} \right) = 1643750 \frac{W}{m^2}$$

$$q''_{tot} = 5.9252 \times 10^{10} \frac{W}{m^2}$$

this is found w/ info given but technically there should be heat loss in air resulting in different  $\Delta T$  but we do not have that information

2.)   $r_1 = 0.1 m$   
 $r_2 = 0.15 m$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left( k r^2 \frac{\partial T}{\partial r} \right) = 0$$

$$k r^2 \frac{\partial T}{\partial r} = C_1 \Rightarrow \frac{\partial T}{\partial r} = \frac{A}{k r^2}$$

$$T(r) = \frac{-A}{k r} + B$$

B.C.'s  $\Rightarrow$  2 needed:  $T(r_1) = 473 = \frac{-A}{k r_1} + B \Rightarrow$

$$T(r_2) = 323 = \frac{-A}{k r_2} + B$$

$$473 - 323 = \frac{A}{k} \left( -\frac{1}{r_1} + \frac{1}{r_2} \right) \Rightarrow \frac{A}{k} = -45 \Rightarrow A = -45 k \Rightarrow$$

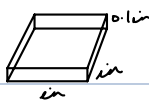
$$473 = \frac{-45}{r_i} + B \Rightarrow B = 473 + \frac{45}{r_i} = 923 \text{ K}$$

$$k \left. \frac{\partial T}{\partial r} \right|_{r=r_i} = \frac{2}{SA} = \frac{6000}{4\pi r_i^2 m^2} \Rightarrow \frac{\partial T}{\partial r} = \frac{A}{kr^2}$$

$$\frac{A}{r_i^2} = \frac{2}{SA} \Rightarrow A = \cancel{r_i^2} \cdot \frac{60}{4\pi \cancel{r_i}} = \frac{60}{4\pi} = 4.77 \text{ W}$$

$$473 = -\frac{4.77}{kr_i} + 923 \Rightarrow \frac{473-923}{4.77} = \frac{1}{kr_i} \Rightarrow -9.424 = \frac{1}{k} \Rightarrow k = 0.106 \frac{\text{W}}{\text{m}\cdot\text{K}}$$

for: parallel for normal

3.)   $\frac{1 \text{ in}}{10 \text{ layers}} = 0.1 \frac{\text{in}}{\text{layer}} \Rightarrow$    

parallel:  $A = \frac{Q}{\frac{k\Delta T}{x}} \Rightarrow 5 \cdot q_{cu} = 5 \left( \frac{Ak\Delta T}{x} \right) \Rightarrow 5q_{cu} = \frac{5(0.1 \text{ in}^2)\Delta T k_1}{1 \text{ in}} \quad \text{eq 1}$ ,  $5q_{glass} = \frac{5(0.1 \text{ in}^2)\Delta T k_2}{1 \text{ in}} \quad \text{eq 2}$

combine eq 1 & eq 2:  $q_{T, \text{parallel}} = 5(0.1 \text{ in}^2)\Delta T(k_1 + k_2)$

normal:  $5 \cdot q_{cu} = 5 \cdot \frac{Ak\Delta T}{x} \Rightarrow 5q_{cu} = \frac{5(1 \text{ in}^2)\Delta T k_1}{0.1 \text{ in}} \quad \text{eq 3}$ ,  $5q_{glass} = \frac{5(1 \text{ in}^2)\Delta T k_2}{0.1 \text{ in}} \quad \text{eq 4}$

combine eq 3 & 4:  $q_{T, \text{normal}} = \frac{5}{0.1 \text{ in}} \Delta T(k_1 + k_2)$  all together  $\Rightarrow$

$\frac{q_{T, \text{parallel}}}{q_{T, \text{normal}}} = \frac{5(0.1) \cancel{\Delta T(k_1 + k_2)}}{(\frac{5}{0.1}) \cancel{\Delta T(k_1 + k_2)}} \Rightarrow \frac{0.5}{50} = 0.01$